



SMART TRACE

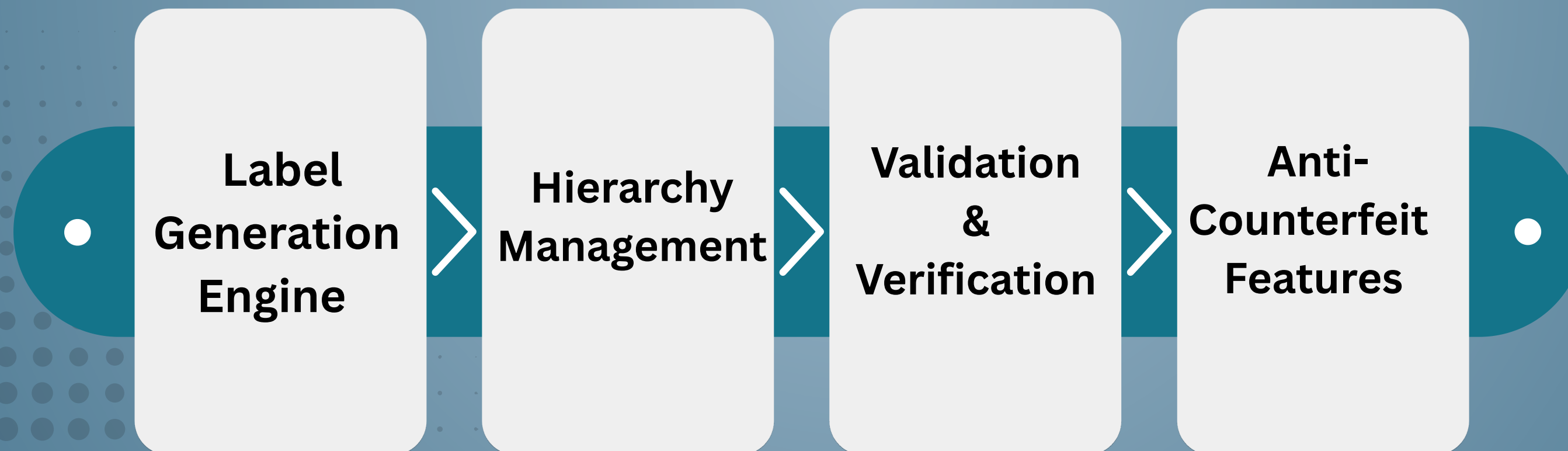
The Digital-Red Thread
of Fate



PROBLEM STATEMENT



A pharmaceutical manufacturer seeks end-to-end serialization to prevent counterfeiting and enable supply-chain traceability, while managing high-volume unique ID generation, packaging hierarchy mapping, and field-level tamper detection.



LABEL GENERATION

×

Generate 3-Tier Batch

Configure the Pallet → Carton → Unit hierarchy.

Product Name

Product Code [Auto-Gen](#)

e.g. Paracetamol

GTIN-141834

Total Units to Produce

1000

AGGREGATION RULES

Units per Carton

Cartons per Pallet

10

50

Generating:

Containing:

Total:

2 Pallets

100 Cartons

1000 Units

Cancel

Start Production

3-Tier Serialization generates GS1-compliant IDs for units, cartons, and pallets in one workflow.

Dynamic Aggregation links parent-child packaging relationships using configurable rules.

Integrity Protection prevents duplicates and detects tampering through counters and check digits.

VALIDATION AND VERIFICATION

Hierarchy Check

Verify unit-to-carton relationships for supply chain traceability

STEP 1: PRODUCT UNIT

GTIN-257053-B16-64-132882A0

⇕

STEP 2: PARENT CARTON

CTN-16-7-31695194

Verify Hierarchy Match



MATCH CONFIRMED

✓ MATCH CONFIRMED: Product found in Carton CTN-16-7-31695194

Hierarchy Check

Verify unit-to-carton relationships for supply chain traceability

STEP 1: PRODUCT UNIT

GTIN-257053-B16-64-132882A0

⇕

STEP 2: PARENT CARTON

CTN-16-7-31695193

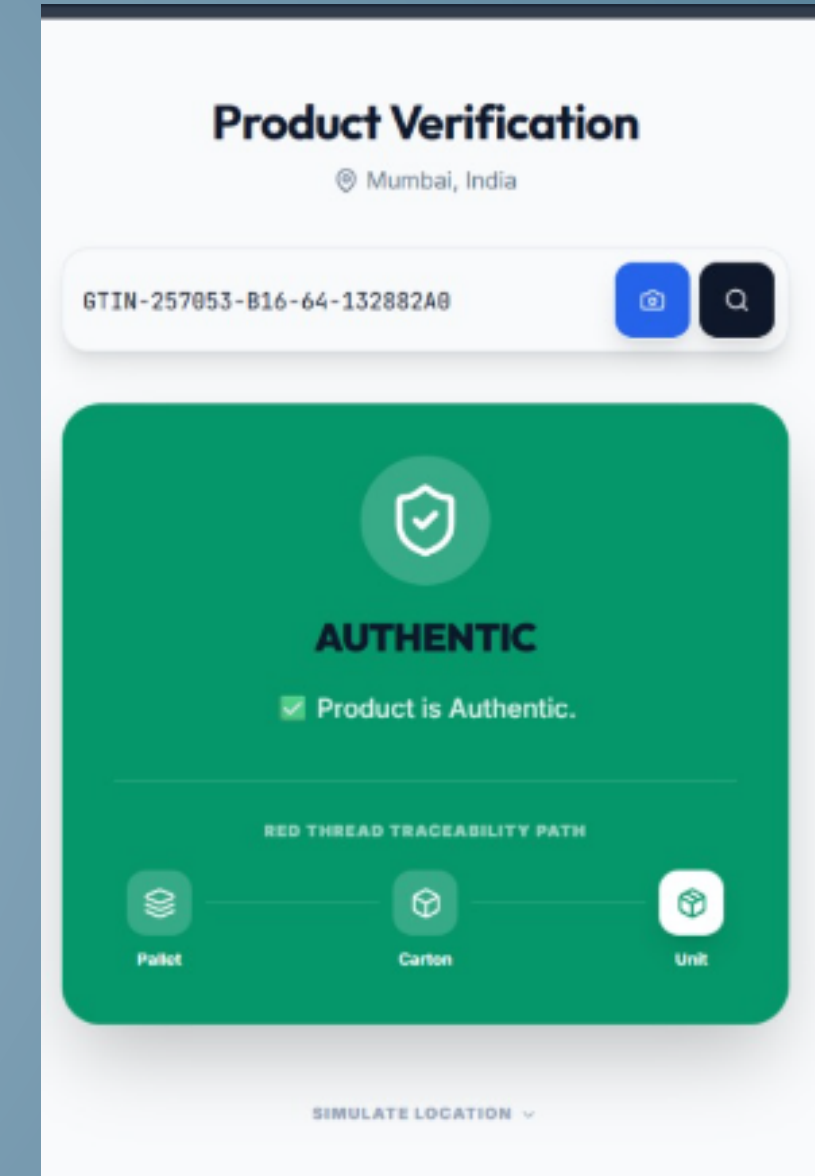
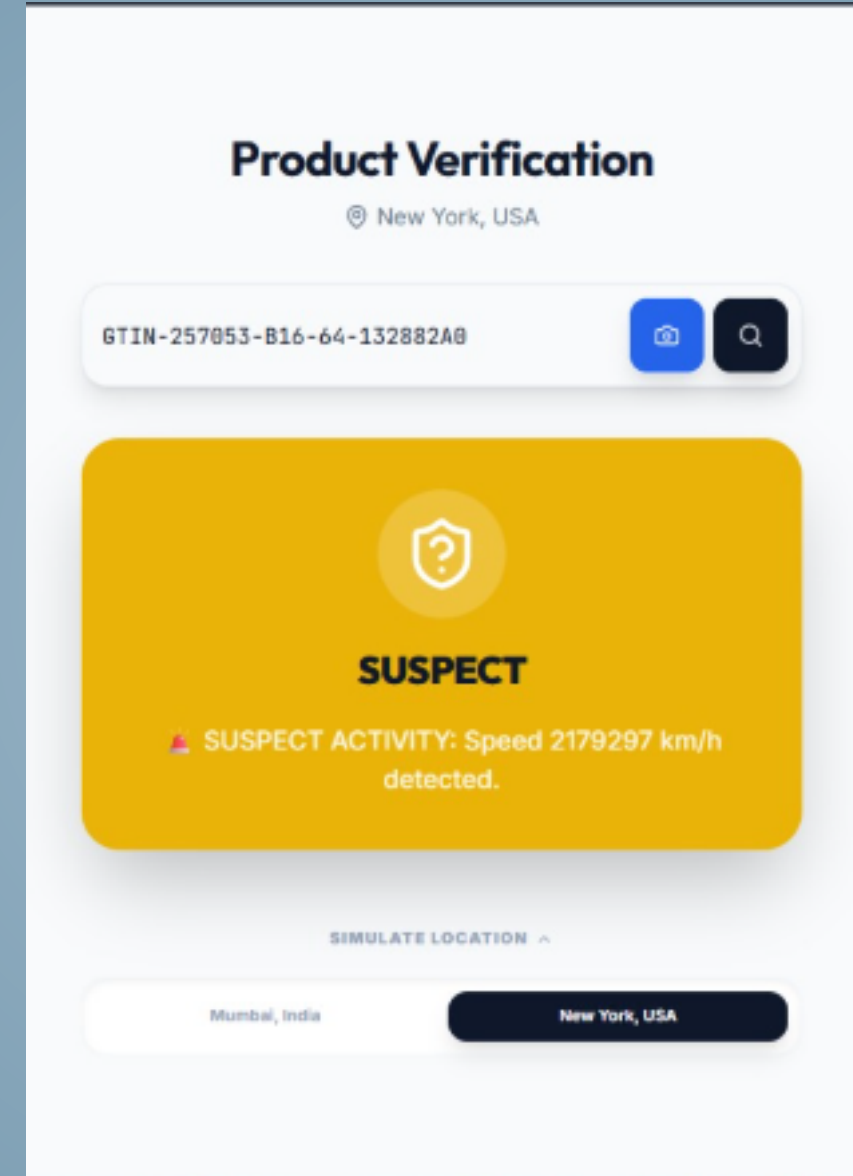
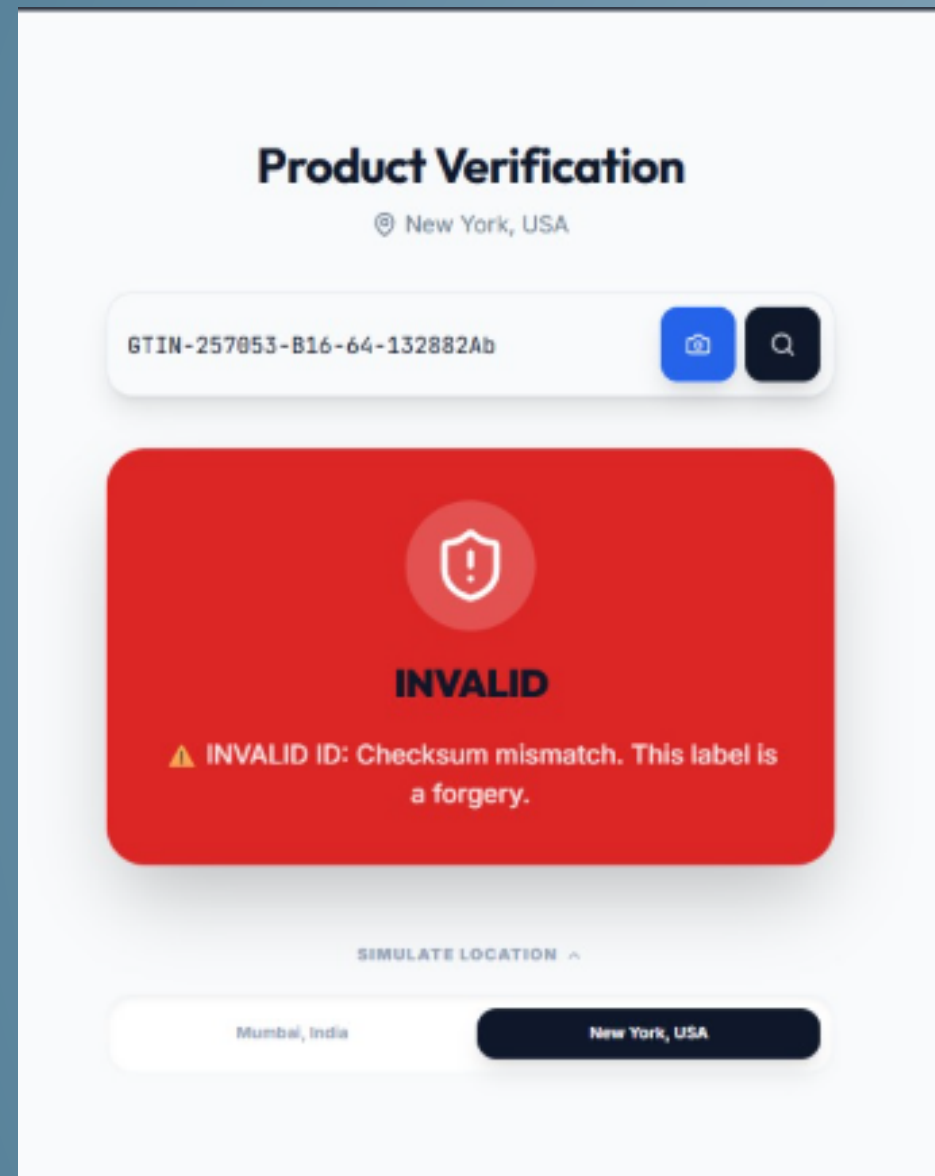
Verify Hierarchy Match



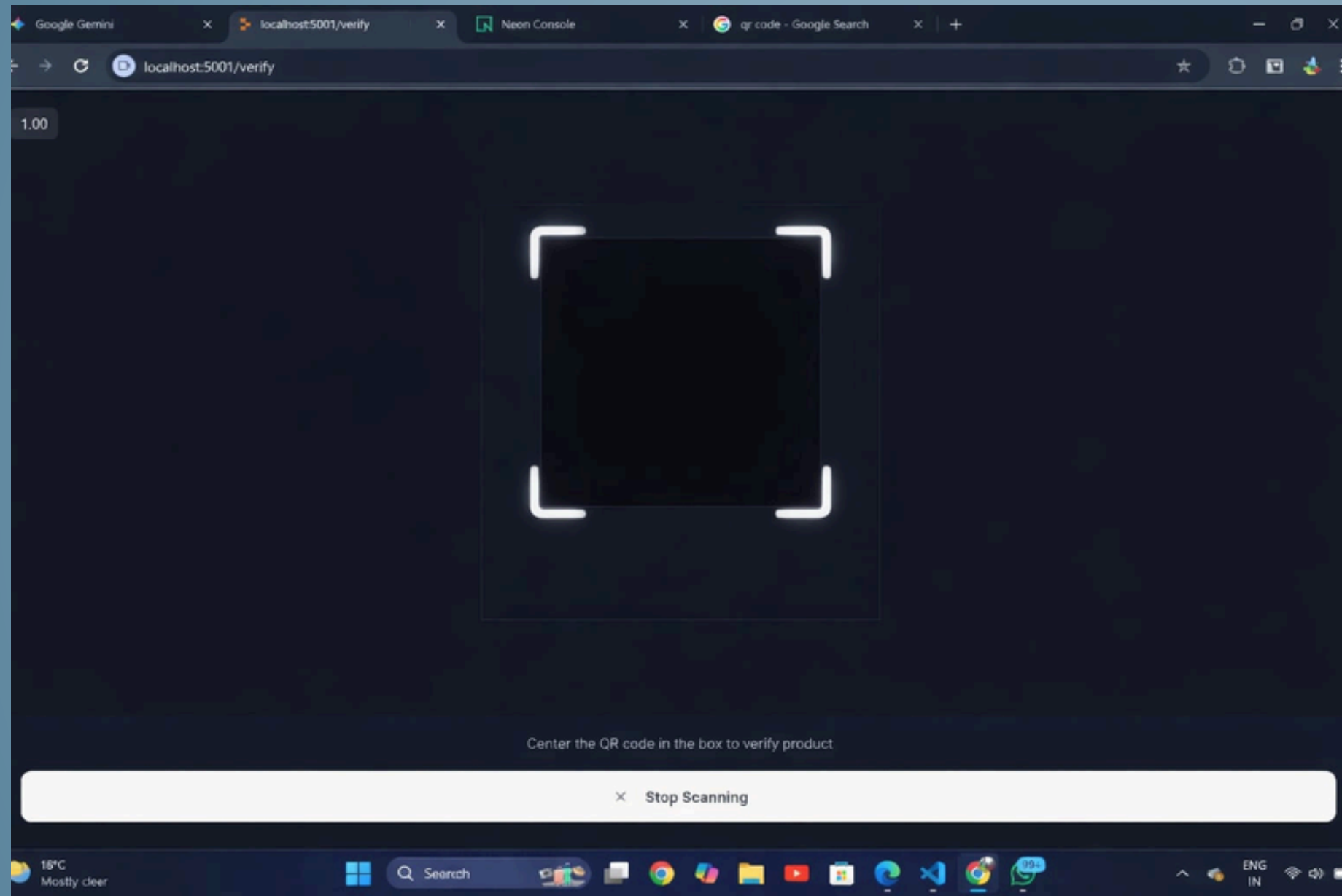
HIERARCHY MISMATCH

Carton ID 'CTN-16-7-31695193' not found.

ANTI-COUNTERFEIT FEATURES



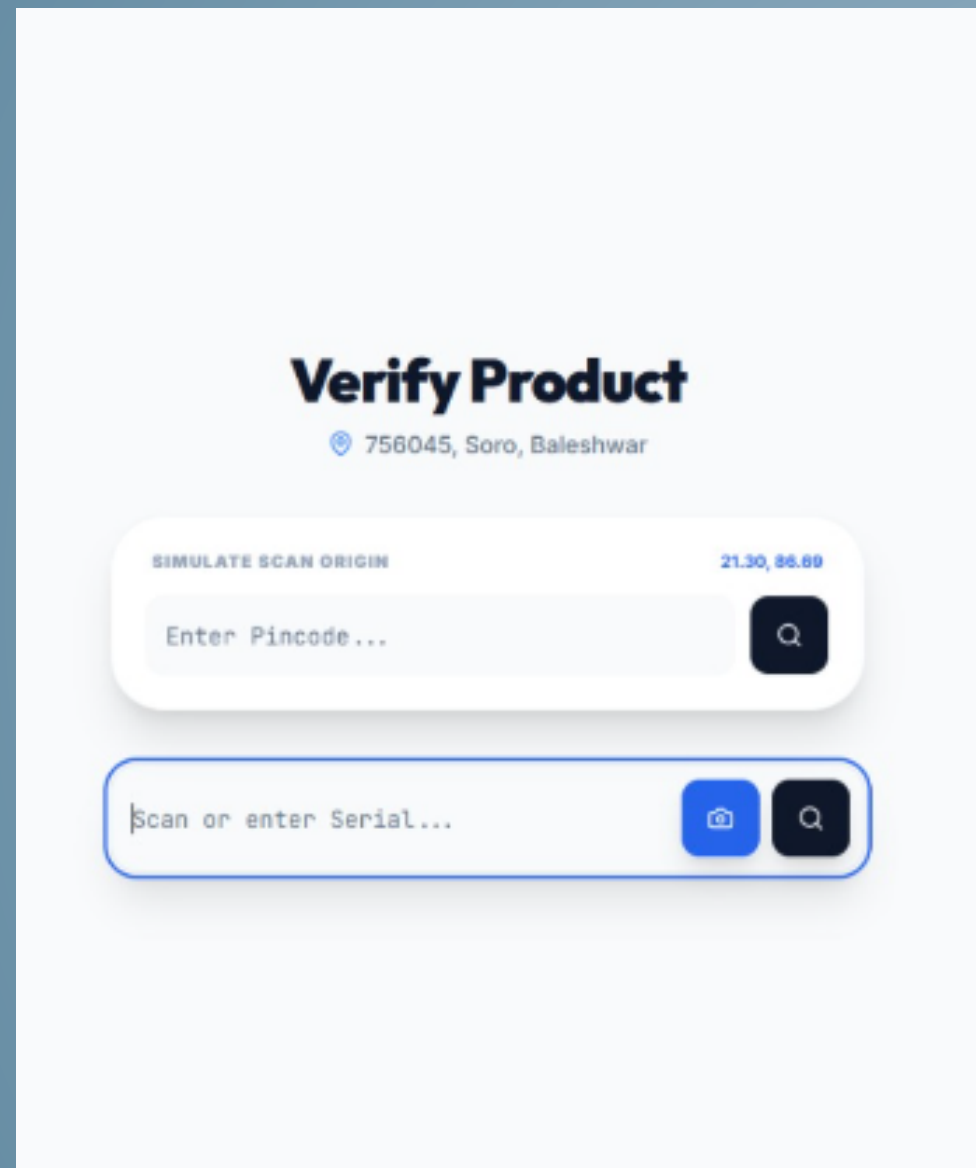
SCANNER INTERFACE



Web interface for the
"Scan-and-Verify"
workflow.



LOCATION VALIDATION



Location-Based Validation verifies scanned serials by comparing the entered pincode's coordinates against expected movement distance thresholds to detect anomalies.



TEAM
The Dank Guys